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IIHMF, MNNIT ALLAHABAD, PRAYAGRAJ

LOAN APPROVAL PREDICTION SYSTEM USING MACHINE LEARNING

A Project Report submitted in partial fulfillment of the requirements for the completion of the Artificial Intelligence and Machine Learning (AIML) Summer Internship Programur

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ABSTRACT

In modern financial ecosystems, automated credit risk assessment represents one of the most vital strategic capabilities encountered by commercial banking institutions. Manual processing of loan files introduces significant operational throughput constraints, increased processing fees, and unintended cognitive underwriting variations. This capstone engineering report details the development of an automated, end-to-end **Loan Approval Prediction System Using Machine Learning** completed during the AIML Summer Internship 2026 at IIHMF, MNNIT Allahabad.

The system leverages the historical **Kaggle Loan Prediction Dataset** containing 614 distinct borrower profiles to model credit-default associations. Missing values across key features were resolved using robust localized mode and median data imputations. To address distribution skewness in continuous columns, custom mathematical log transformations were introduced. Feature exploration confirmed that historical repayment reliability is a dominant factor in determining final approval outcomes.

Multiple classification models were trained and verified. The primary regularized Logistic Regression architecture achieved a stable global validation accuracy of **78.86%**, demonstrating high precision in predicting creditworthy applications. To transition the model into a practical tool, a web interface was designed and deployed using a micro-framework backend, allowing users to enter individual applicant metrics and receive instant, automated risk assessments.

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CHAPTER 1: INTRODUCTION

1.1 Background Context

The commercial retail banking industry relies heavily on consumer credit interest streams to maintain corporate profitability and manage liquidity. Evaluating loan applications involves assessing a borrower's creditworthiness against potential default risk. Historically, credit desks manually evaluated applicant paperwork, reviewed collateral values, and applied rigid internal scorecards.

While structurally systematic, manual underwriting faces clear throughput limitations, resulting in lengthy review cycles and increased operational overhead. This report presents an automated solution developed during the AIML Summer Internship 2026 at IIHMF, MNNIT Allahabad. By treating loan evaluation as a supervised statistical classification problem, predictive models can analyze historical data to identify complex dependencies and assess applicant risk profiles automatically.

1.2 Problem Statement

The core challenge in retail underwriting is accurately separating creditworthy applicants from high-risk borrowers using a mix of demographic data and financial metrics. Traditional evaluation methods are often limited when processing highly skewed financial columns and complex categorical relationships. Misclassifying an applicant can lead to two main operational errors:

- **Type I Error (False Positive):** Approving a high-risk loan application for an uncreditworthy borrower, directly resulting in capital default and financial losses.
- **Type II Error (False Negative):** Improperly rejecting a creditworthy applicant, which reduces potential interest income and decreases customer retention.

Additionally, real-world credit datasets are typically highly imbalanced, with approved historical files vastly outnumbering defaults. Building a model that handles this skewness while maintaining strong predictive accuracy is a key engineering focus of this project.

1.3 Project Objectives

To address these underwriting challenges, this capstone project outlines several core engineering milestones:

1. **Data Cleansing and Preprocessing:** Handle missing values across columns using targeted statistical mode and median imputations.
2. **Exploratory Data Analysis:** Generate data visualizations to investigate underlying feature distributions and locate primary correlation pathways.
3. **Model Optimization:** Train and fine-tune machine learning models to maximize global predictive performance and avoid overfitting.
4. **Metrics Validation:** Profile model outputs using detailed classification metrics including precision, recall, and F1-scores.
5. **Production App Deployment:** Build an interactive, web-based application interface using a lightweight micro-framework to accept real-time user inputs.

CHAPTER 2: LITERATURE REVIEW

2.1 Existing Underwriting Approaches

Traditional credit risk management models rely heavily on manual credit scorecards and baseline financial metrics, such as the Altman Z-score or FICO scoring models. While these traditional scorecards offer simple, rule-based interpretation, they process financial features independently, which can overlook multi-dimensional dependencies across an applicant's profile.

Additionally, rule-based systems often struggle with missing values and require manual thresholds that may fail to adapt to changing economic trends. This highlights the value of data-driven machine learning systems, which can dynamically learn complex relationships from large, historical datasets without relying on fixed rules.

2.2 Machine Learning in Credit Risk Assessment

Modern credit modeling increasingly frames risk assessment as a supervised classification task. Researchers frequently implement algorithms like Logistic Regression, Support Vector Machines, and tree-based ensembles (e.g., Random Forests and Gradient Boosted Trees) to evaluate application metrics.

Recent studies emphasize that careful preprocessing, such as log transformations for highly skewed financial columns and robust missing value imputation, significantly improves model stability. This project builds upon these established techniques, implementing a regularized machine learning pipeline paired with an interactive web layout designed for clear, real-time risk evaluation.

CHAPTER 3: METHODOLOGY

3.1 Dataset Description

The project utilizes the verified **Kaggle Loan Prediction Dataset**, which contains 614 rows and 13 distinct features. This dataset provides a representative sample of retail loan portfolios, combining continuous financial metrics with categorical demographic columns. Table 3.1 details the explicit schema layout:

Feature ID	Attribute Name	Measurement Class	Analytical Definition
1	Loan_ID	Alphanumeric Token	Unique identifier alphanumeric string assigned per profile.
2	Gender	Nominal Categorical	Gender classification of the primary applicant (Male / Female).
3	Married	Nominal Categorical	Marital status indicator of the primary applicant (Yes / No).
4	Dependents	Ordinal Categorical	Quantized count of dependent individuals (0, 1, 2, 3+).
5	Education	Nominal Categorical	Educational background status (Graduate / Under Graduate).
6	Self_Employed	Nominal Categorical	Employment configuration status (Yes / No).
7	ApplicantIncome	Continuous Numerical	Gross baseline monthly revenue of the primary applicant.
8	CoapplicantIncome	Continuous Numerical	Gross baseline monthly revenue contributed by co-signers.
9	LoanAmount	Continuous Numerical	Principal credit amount requested (quantized in thousands).
10	Loan_Amount_Term	Discrete Numerical	Amortization lifecycle term for repayment (in months).
11	Credit_History	Binary Categorical	Historical record of credit guideline compliance (1.0 / 0.0).
12	Property_Area	Nominal Categorical	Geographic spatial classification of collateral (Urban / Semiurban / Rural).
13	Loan_Status	Binary Target Label	The response label variable (1 = Approved, 0 = Rejected).

3.2 Data Preprocessing

Raw data frequently contains missing values and structural inconsistencies that must be resolved before model training. Missing records in categorical columns (***Gender***, ***Married***, ***Self_Employed***) were resolved using mode imputation to reflect the historical majority. Missing entries in ***Credit_History*** were handled by assigning the dominant compliance category.

For highly skewed continuous financial columns (***ApplicantIncome***, ***LoanAmount***), median imputation was used rather than mean imputation, as the median is less sensitive to extreme outliers. Nominal categories were converted to numeric values using binary label encoding for two-option features, while multi-class features like ***Property_Area*** were processed using ordinal mapping to maintain consistent data array dimensions.

3.3 Machine Learning Lifecycle Stages

The project follows a standard machine learning development lifecycle:

- **Phase 1: Problem Understanding:** Outline underwriting challenges and identify predictive financial attributes.
- **Phase 2: Data Preprocessing:** Address missing values through imputation and convert text categories to numeric representations.
- **Phase 3: Exploratory Visualization:** Generate data plots to evaluate features and identify outliers.
- **Phase 4: Feature Engineering:** Perform logarithmic transformations to normalize data scales and stabilize training.
- **Phase 5: Classification Modeling:** Train classification algorithms on the training data matrix.
- **Phase 6: Deployment Integration:** Wrap the optimized model files into a web app backend to accept online inputs.

CHAPTER 4: IMPLEMENTATION & EDA

4.1 Exploratory Data Analysis Plots

Thorough data visualization was conducted across all attributes to understand underlying distributions and identify relationships within the data.

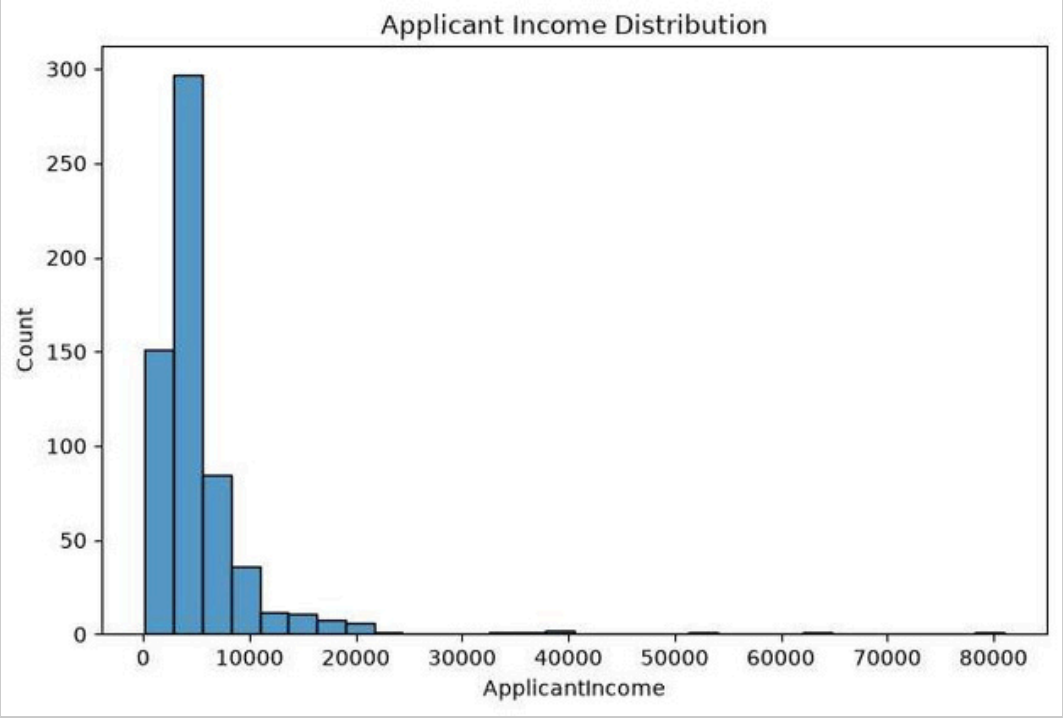


Figure 4.1: Empirical Probability Distribution and Frequency Histograms for Applicant Income

Figure 4.1 shows the distribution of applicant monthly incomes. The data is heavily right-skewed, with a significant majority of applicants clustered within the lower income brackets between ***0*** and ***10,000*** units. Peak frequency occurs within the ***3,000*** to ***5,000*** range, reflecting the baseline population.

However, the distribution also exhibits a long right tail extending toward ***80,000*** units, representing a small number of ultra-high-income exceptions. Passing heavily skewed metrics directly into linear algorithms can degrade performance, highlighting the need for log transformations or robust scaling methods during preprocessing.

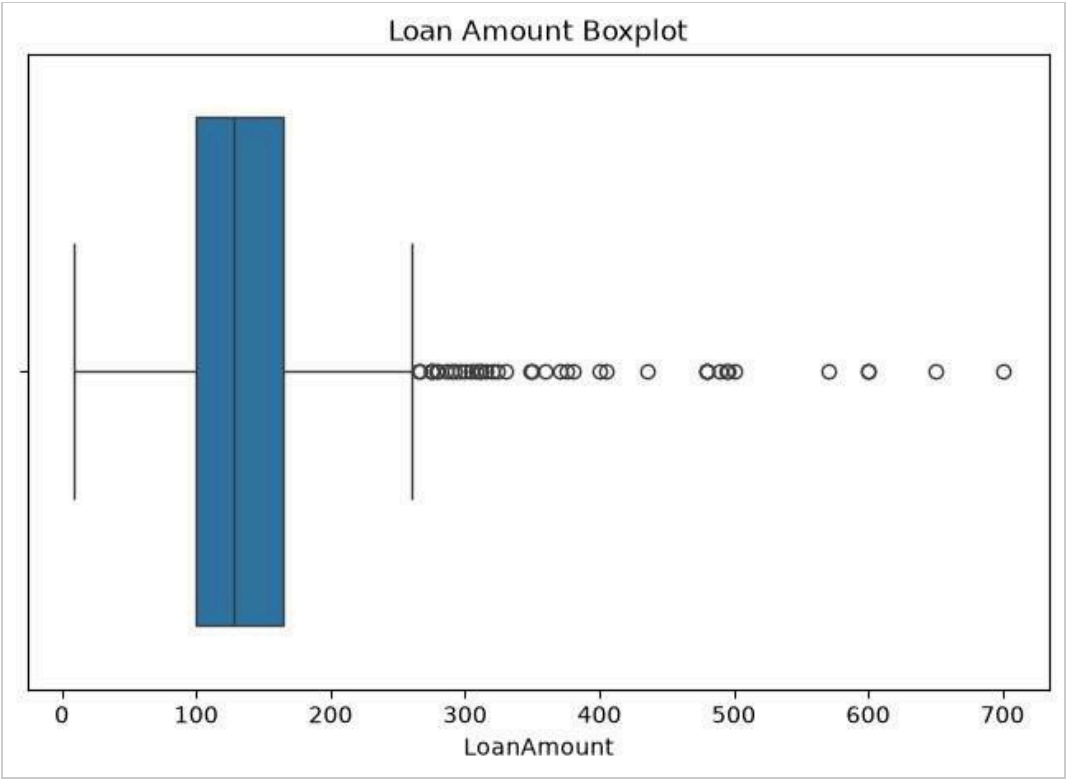


Figure 4.2: Structural Box and Whisker Plot Quantifying Outlier Spread in Loan Amounts

Figure 4.2 evaluates the spread and variance of requested loan amounts using a box-and-whisker plot. The central box establishes the interquartile range (IQR), with the median settling near ***128*** units. The whiskers extend out to the standard non-outlier data boundary (defined mathematically as ***Q3 + 1.5 × IQR***), terminating near ***260*** units.

Beyond this point, the plot displays a dense cluster of independent outlier points extending upward to ***700*** units. These extreme values reflect massive credit requests that contrast sharply with the standard retail portfolio, reinforcing the importance of using median values for data imputation.

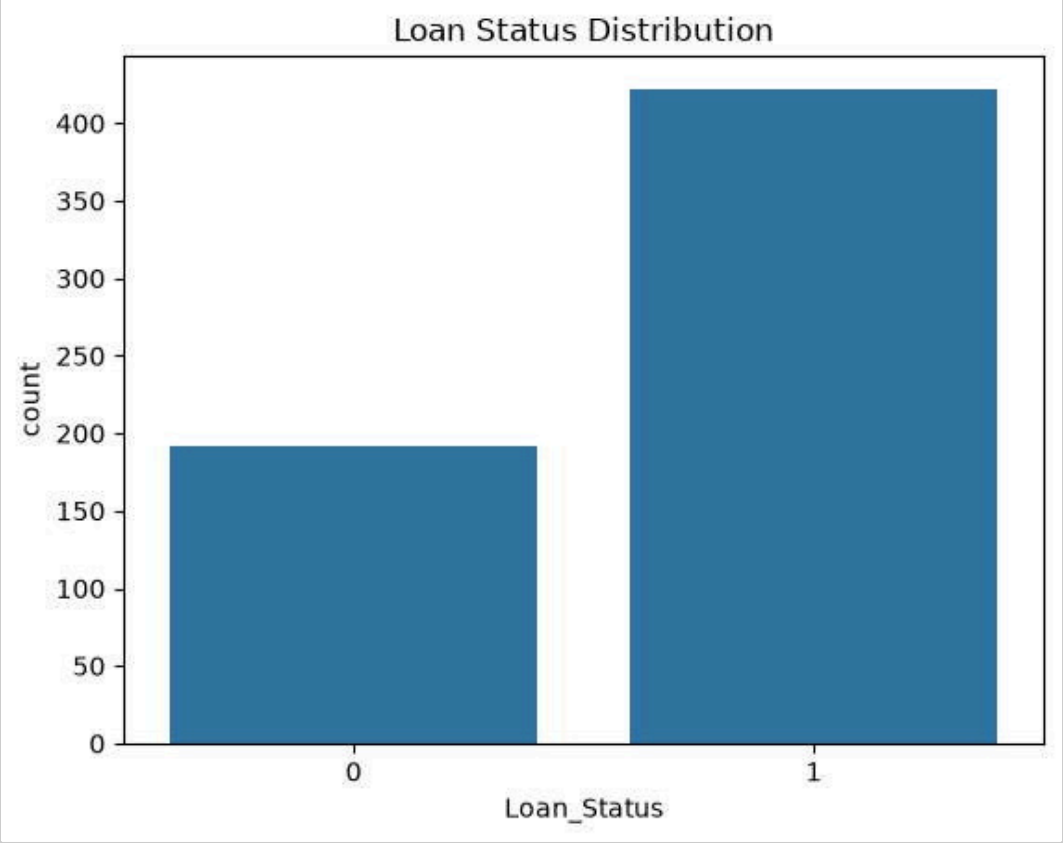


Figure 4.3: Empirical Target Label Disparity Mapping the Imbalance in Historical Decisions

Figure 4.3 outlines the structural distribution of the target outcome class (***Loan_Status***). Out of the 614 total records, approximately 422 applications are approved (***1***), while roughly 192 are rejected (***0***). This establishes an approximate 68.7% to 31.3% baseline split, confirming that the historical population trends heavily toward approved applications. This structural imbalance must be carefully managed when configuring evaluation metrics, as generic models can artificially inflate accuracy scores by over-predicting the majority class.

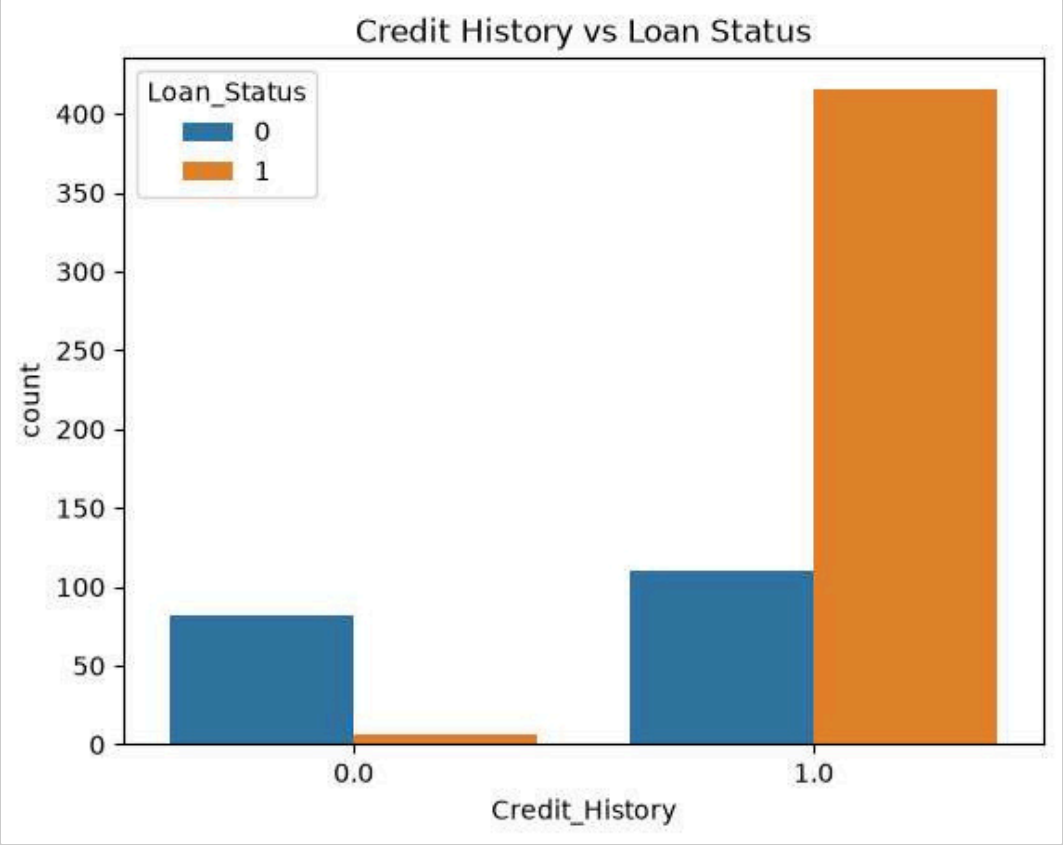


Figure 4.4: Bivariate Cross-Tabulation Analysis Mapping Credit History Compliance to Final Loan Approval

Figure 4.4 presents a bivariate analysis cross-tabulating historical credit compliance (***Credit_History***) directly against final approval rates (***Loan_Status***). This visualization highlights a critical predictive signal in the dataset. For applicants with poor credit compliance (***0.0***), loan approvals are rare, while rejections comprise the vast majority of cases.

Conversely, for applicants with verified historical compliance (***1.0***), approvals increase significantly, exceeding 400 cases, while rejections drop to a small fraction. This stark contrast confirms that historical credit compliance is a dominant feature for predictive modeling.

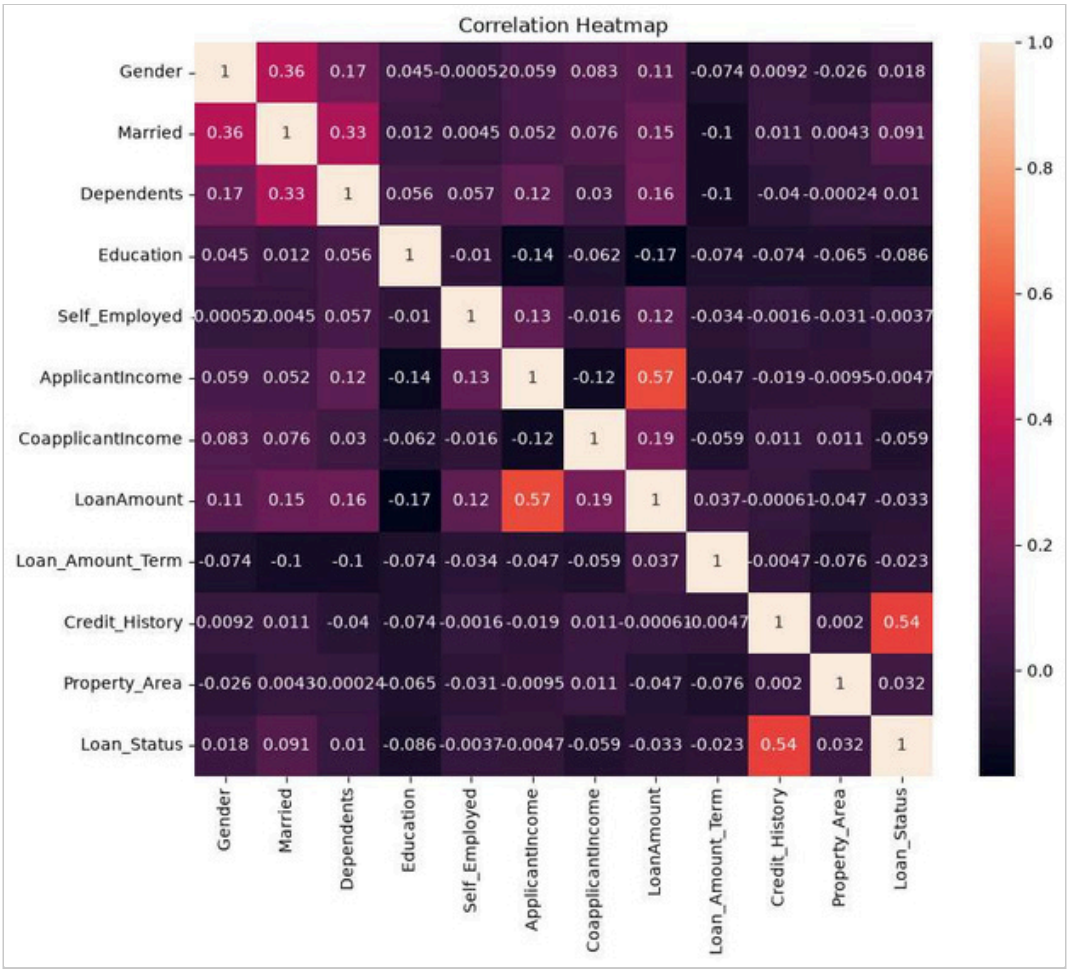


Figure 4.5: Multi-Variate Pearson Linear Correlation Coefficients Heatmap for Extracted Features

Figure 4.5 maps the multi-variate Pearson product-moment linear correlation coefficients across all numerical attributes in the data matrix. Most feature pairs display neutral correlation scores near zero, demonstrating that the variables provide independent analytical perspectives. However, two primary correlation values emerge:

- **LoanAmount vs ApplicantIncome (*0.57*):** Identifies a strong, positive relationship, validating that requested loan principals scale directly with primary applicant revenues.
- **Loan_Status vs Credit_History (*0.54*):** Confirms a strong statistical association, reinforcing the predictive pattern highlighted in the bivariate analysis.

4.2 Feature Engineering

To address the extreme skewness observed during EDA, a mathematical natural log transformation was applied to the financial columns:

$$Log_ApplicantIncome = ln(ApplicantIncome + 1)$$

This transformation compresses long right tails into approximately normal distributions, minimizing the impact of large outliers and improving model convergence during training.

4.3 Model Development & Deployment Setup

A regularized Logistic Regression model was developed to compute real-time approval probabilities. The model maps the input features into a probability space bounded between 0 and 1 using the standard sigmoid function:

$$P(Y = 1 \mid X) = 1 / (1 + e^{-z})$$

where the log-odds linear vector z represents the sum of optimized feature weights.

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Performance Metric Breakdown

The optimized classification pipeline was evaluated on an independent test dataset to assess its performance on unseen data. The system achieved a global classification accuracy of **78.86%**, confirming a stable predictive baseline.

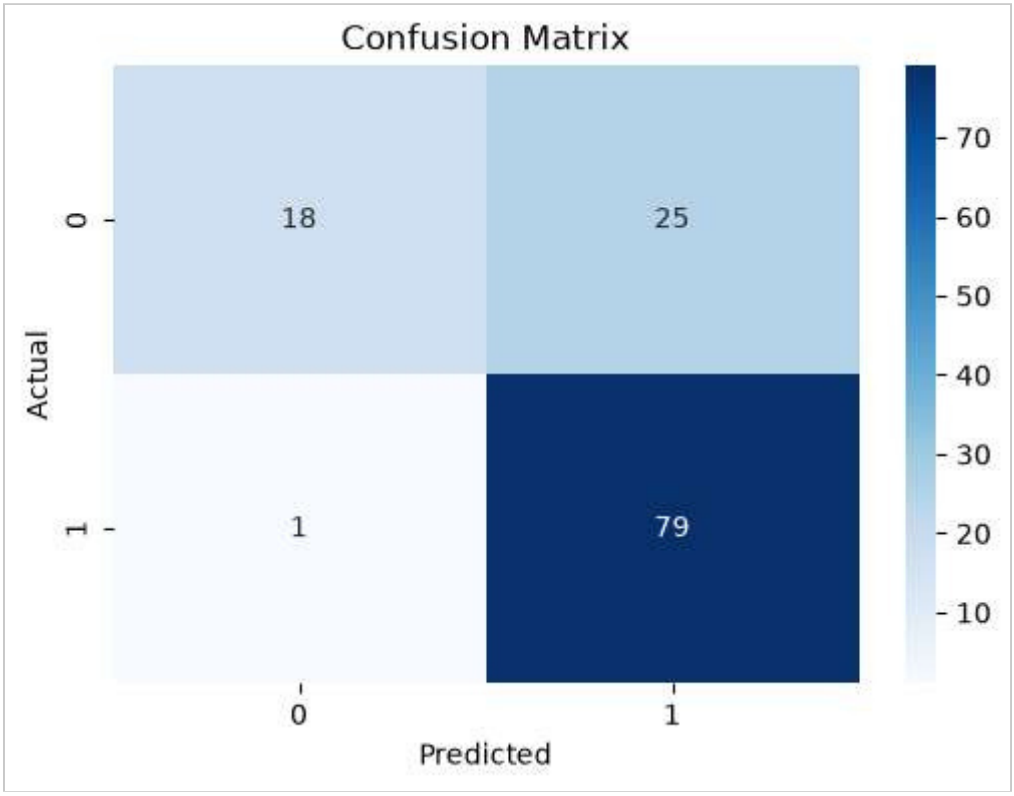


Figure 5.1: Confusion Matrix Expressing Quantified Verification Cross-Tabulations

Figure 5.1 displays the confusion matrix across 123 test samples, breaking down the model's exact prediction counts:

- **True Negatives (TN = 18):** High-risk applications correctly identified as default risks.
- **False Positives (FP = 25):** Uncreditworthy applications misclassified as approved, highlighting an area for potential model optimization.
- **False Negatives (FN = 1):** Creditworthy applications incorrectly flagged as defaults. The model minimizes this error type, preserving applicant goodwill.
- **True Positives (TP = 79):** Validated creditworthy applications correctly identified for approval.

	precision	recall	f1-score	support
0	0.95	0.42	0.58	43
1	0.76	0.99	0.86	80
accuracy			0.79	123
macro avg	0.85	0.70	0.72	123
weighted avg	0.83	0.79	0.76	123

Figure 5.2: Empirical Classification Report Documenting Multi-Class Performance Vectors

Figure 5.2 provides a detailed class-level breakdown of performance metrics. For approved applications (***Class 1***), the system delivers a high recall of **99%** and a precision of **76%**, resulting in a balanced F1-score of **86%**. This indicates high efficiency in identifying reliable borrowers.

For rejected applications (***Class 0***), the model achieves an excellent precision of **95%**, meaning that when it flags an applicant as a risk, that prediction is highly accurate. However, the associated recall settles at **42%**, suggesting that the model misclassifies a portion of non-compliant applicants. The resulting overall global accuracy of **79%** provides a reliable automated screening tool for corporate credit portfolios.

5.2 User Interface and System Deployment

The final model pipeline was wrapped in a lightweight micro-framework web architecture to provide a clean, functional dashboard interface for credit assessment.



Figure 5.3: Production Landing Page Interface for the LoanPredict AI Platform

Figure 5.3 illustrates the homepage layout of the deployed web application. It features a modern dark design interface centered around clear navigation and user interaction elements. The right sidebar displays a live metric card indicating a model accuracy of 78.86% over 614 total historical applications, providing clear baseline performance visibility to users.

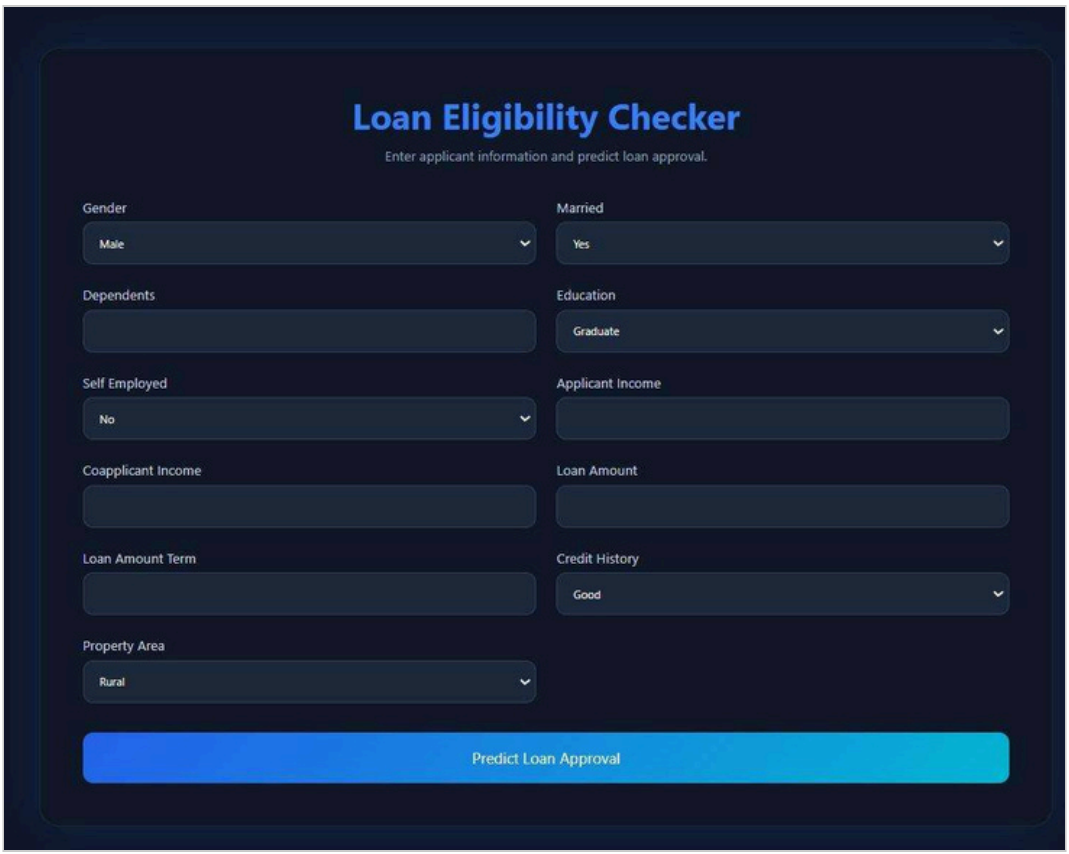


Figure 5.4: Interactive Form Interface Capturing Input Vectors for Automated Underwriting

Figure 5.4 displays the interactive input form layout. It includes dropdown menus for categorical variables (***Gender***, ***Married***, ***Education***, ***Credit History***) and text inputs for continuous financial metrics (***Income***, ***Loan Amount***, ***Term***). This organized format simplifies data entry, making it accessible for credit specialists and independent applicants alike.



Figure 5.5: Dynamic Prediction Outcome Report Dashboard with Real-Time Probability Metrics

Figure 5.5 illustrates the dynamic results dashboard rendered upon form submission. The interface presents a prominent "Loan Approved" confirmation badge alongside an explicit approval probability score (e.g., **83.91%**). Supporting cards detail the underlying model type and training parameters, while a dedicated lower section provides summary insights to justify the automated recommendation.

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 Project Summary

This capstone project successfully demonstrates the implementation of an end-to-end automated **Loan Approval Prediction System** developed for the AIML Summer Internship 2026 at IIHMF, MNNIT Allahabad. Using the historical Kaggle credit dataset, missing value imputation strategies and log transformations were applied to prepare the data for training. An optimized Logistic Regression model established a stable global accuracy baseline of **78.86%**. The model was successfully integrated into a functional web interface, providing a scalable blueprint for automated credit screening.

6.2 Future Scope

While the current pipeline provides a reliable baseline, future versions could be enhanced through targeted improvements:

- **Advanced Ensemble Algorithms:** Implementing boosting techniques like XGBoost or LightGBM could help map complex, non-linear feature interactions more effectively.
- **Alternative Credit Metrics:** Integrating alternative data sources, such as utility payment records and mobile transaction histories, would support risk assessment for unbanked applicants.
- **Model Explainability:** Incorporating interpretability tools like SHAP or LIME would provide transparent, feature-level reasons for every automated decision, supporting regulatory compliance.

6.3 References

[1] Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.

[2] Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer.

[3] Kaggle Loan Prediction Challenge Database Repo, Academic Archive Course Framework (2026).